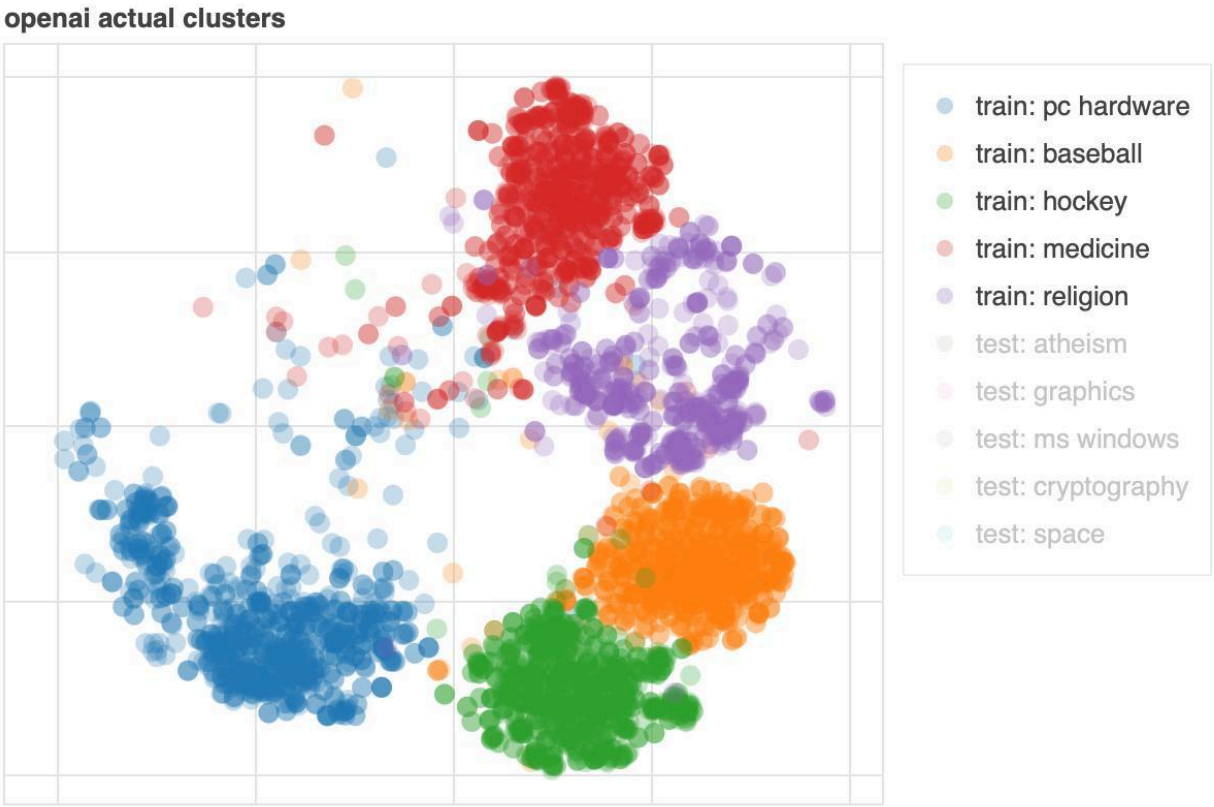
Practical Assignment 5: Intelligent Document Search Using Unsupervised Learning



Application: Build an intelligent search system for research papers or college notes.

Problem statement: Design a system that automatically discovers groups of related documents, identifies their topics, and retrieves useful documents without requiring manually labeled training data.

Tasks to perform

1. Collect documents: Gather at least 20 documents from three or more subject areas, such as AI, cybersecurity, databases, and networking.
2. Generate word embeddings: Represent words or documents as numerical vectors using Word2Vec, GloVe, or a sentence-embedding model.

3. Apply clustering: Use K-means or hierarchical clustering to group similar documents automatically.
4. Perform topic modeling: Apply Latent Dirichlet Allocation (LDA) or another suitable topic-modeling method to discover the main themes.
5. Reduce dimensionality: Use PCA or t-SNE to visualize documents in two dimensions and observe their clusters.
6. Detect anomalies: Identify documents that do not fit well into any cluster, such as an unrelated document accidentally added to a collection.
7. Build a retrieval interface: Allow users to enter a query and retrieve documents based on semantic similarity or cluster membership.
8. Evaluate retrieval quality: Create a small set of relevance judgments and calculate Precision, Recall, or another suitable metric.

Example scenario

Suppose students upload the following documents:

Document	Main subject
D1	Neural networks and deep learning
D2	Natural language processing
D3	Network security and firewalls
D4	Cryptography and cyberattacks
D5	Relational databases and SQL
D6	Database indexing

D7	A document about gardening

Expected clustering:

- Cluster 1: AI and NLP — D1, D2
- Cluster 2: Cybersecurity — D3, D4
- Cluster 3: Databases — D5, D6
- Potential anomaly: D7, if it is sufficiently dissimilar to the other documents.

Query: “documents about protecting computer networks”

The system should retrieve D3 and D4 based on their content, even if the exact query words do not appear in every document.

Expected output

- Document embeddings or vector representations.
- Automatically generated document clusters.
- Topic summaries for each cluster.
- A 2D visualization of the document collection.
- A list of detected anomalies.
- Search results with an explanation of their relevance.

Concepts covered: Word embeddings, clustering-based retrieval, topic modeling, dimensionality reduction, anomaly detection, and retrieval evaluation.